(Please write your Enrollment Number) Enrollment No. 12701180203

MID-TERM EXAMINATION

(Course Name: B. Tech (ECE) 2nd Year) (Semester: 3rd)

(October 2024) OFFLINE mode

Subject Code: BEC 205	Subject: Network Analysis and Synthesis
Time: 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Define Kirchhoff's voltage law.		
	b) Derive the current division rule.		
	c) Explain the process of source transformation.		
	d) Find Laplace Transform of e^{at}.		
	UNIT I		CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) For the circuit shown in fig-1, determine the current (I) through the 10 Ω resistance using Superposition Theorem.		
	b) For the circuit shown in fig-1, determine the current (I) through the 10 Ω resistance using Thevenin's Theorem		
	c) For the circuit shown in fig-1, determine the current (I) through the 10 Ω resistance using Mesh Analysis Method		
	UNIT II		CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) Consider the RC series circuit shown in Fig. 2 (Vs = 2V), (R = 6 Ω), (L = 2 H), (C = 0.25 F). Determine i(0) , di(0)/dt, $d^2i(0)/dt^2$ and i(t).		
	b) Draw the graph of network shown in fig. 3 and also write Incidence Matrix for given network.		
	c)Find the Laplace transform of $e^{-at}(\sin \omega t + \cos \omega t)$.		

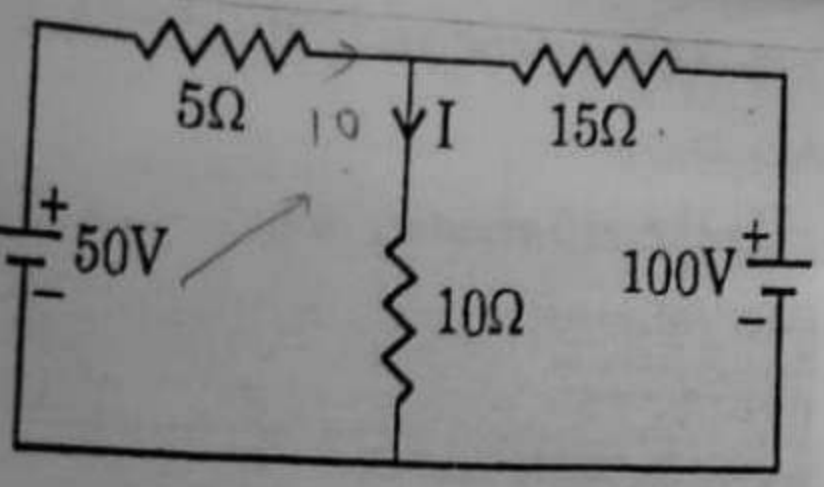


Fig. 1

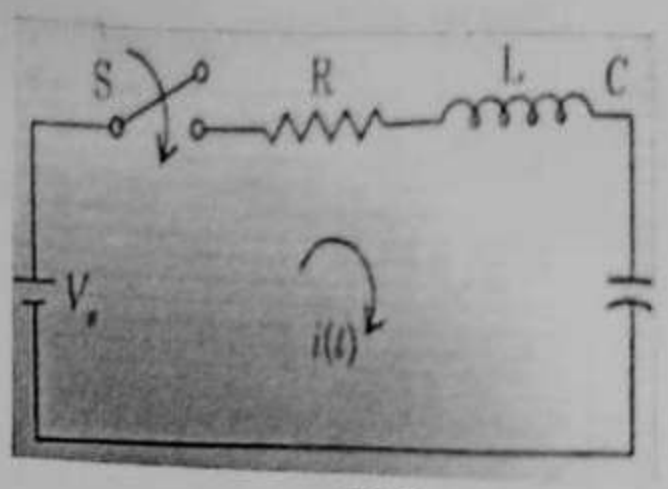


fig.2

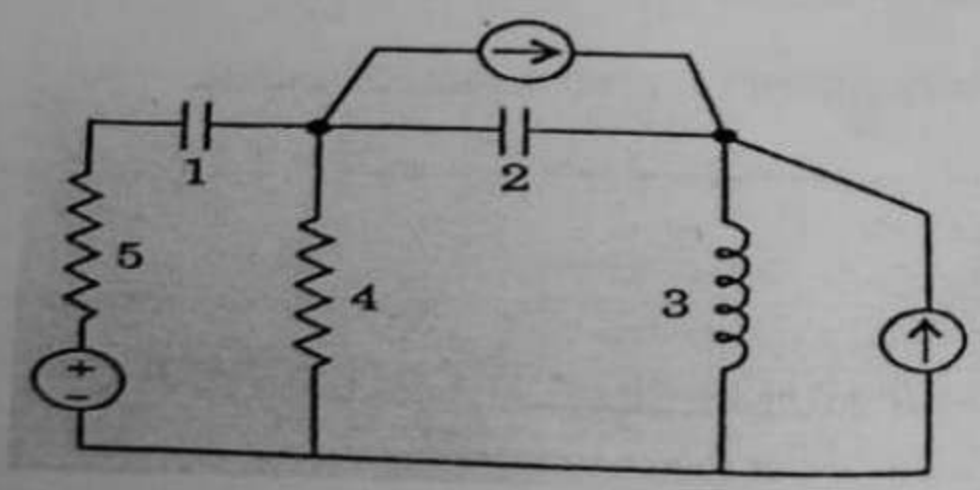


Fig. 3